

Resolution Geometry of Induced Observable Transport Levi-Civita Closure, Projected Sector Connections, Entropy No-Go, and the Rank-Change Frontier

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Abstract

Resolution Geometry (RG) previously characterized the observational quotient, local distinguishability metric, hard spectral projector, maximal admissible sector, and compatible transport class, but left the choice of a connection open. We close that problem on every regular observable stratum under a precise tangent-bundle hypothesis. If the characterized observable metric is g , the fundamental theorem of Riemannian geometry fixes the unique torsion-free metric connection ∇^{LC} . If the maximal admissible sector $W_{\text{obs|adm}}$ is a smooth constant-rank g -orthogonal subbundle of the observable tangent bundle with characterized projector P , then

$$\nabla_X^{W_{\text{obs|adm}}} s = P \nabla_X^{\text{LC}} s$$

is a canonical induced metric connection. It is natural under isometries preserving P , reduces to the restricted Levi-Civita connection when P is parallel, and yields the canonical decomposition

$$\nabla_X^{\text{LC}} s = \nabla_X^{W_{\text{obs|adm}}} s + \Pi(X, s), \quad \Pi(X, s) = (I - P) \nabla_X^{\text{LC}} s.$$

Thus local transport is determined by the already characterized pair (g, P) ; no variational entropy selector is required. We also prove a no-go theorem: $H(Q)$ and $H(X \mid Q)$ are invariant under measure-preserving fibre automorphisms and therefore cannot choose a unique connection. Curvature and holonomy are downstream invariants of the induced connection, not generic selectors of it.

A deterministic numerical attack verifies metric compatibility, torsion freedom, projector identities, induced-bundle metric compatibility, Gauss decomposition, affine-coordinate naturality, exact restriction on a parallel sector, entropy invariance across all 36 fibre automorphisms, and explicit failure of the smooth-bundle hypothesis at rank-changing threshold crossings. The strongest conclusion is conditional but exact: regular tangent and admissible-sector connection selection is closed whenever the characterized sector is a smooth orthogonal tangent subbundle. Rank-changing stratified extension, arbitrary non-tangent vector bundles, quantum metric selection, and physical interpretation remain open modules.

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1 Problem and closure statement

The earlier RG representation theorem gives, on a regular fibre,

$$(\mathcal{E}, \Gamma) \mapsto (Q_{\mathcal{E}}, g, P, W_{\text{obs|adm}}, \{T_{ij}\}).$$

The maps T_{ij} were required to be natural, metric and projector compatible, sector preserving, and cocycle closed. Those requirements specify an admissible class but not a unique infinitesimal connection. Entropy-based synchronization attacks sharpened the issue: entropy is useful for excluding quotient-incompatible maps, but it is constant on unresolved fibre automorphisms and cannot remove their stabilizer freedom.

The missing observation is geometric. The connection question is not one question until its bundle is stated. On the tangent bundle of a Riemannian observable stratum, metric compatibility plus vanishing torsion invokes a classical uniqueness theorem. On a smooth orthogonal tangent subbundle, the characterized projector induces a connection by projection. On an arbitrary vector bundle, neither result applies without additional structure.

Main closure. On each regular observable stratum S for which $W_{\text{obs|adm}} \subset TS$ is a smooth constant-rank orthogonal subbundle,

$$(\mathcal{E}, \Gamma) \rightarrow (S, g, P) \rightarrow \nabla^{\text{LC}} \rightarrow \nabla^{W_{\text{obs|adm}}} = P\nabla^{\text{LC}}.$$

The claim is fibrewise and conditional on the stated regularity and tangent-bundle realization. It is not a protocol-free derivation of g , P , or the physical meaning of transport.

2 Established mathematical grounding

Table 1: Established components and their precise RG role.

Result	Established force	RG role
Fundamental theorem of Riemannian geometry	A Riemannian metric determines a unique torsion-free metric connection.	Fixes ∇^{LC} on each regular observable stratum.
Orthogonal projection of a connection	A connection on an ambient metric bundle induces a metric connection on a smooth orthogonal subbundle.	Fixes $\nabla^{W_{\text{obs adm}}} = P\nabla^{\text{LC}}$ once (g, P) are fixed.
Gauss decomposition	Ambient derivative splits into tangential and normal components.	Produces intrinsic admissible transport and the second fundamental form.
Ambrose–Singer holonomy theorem	Curvature generates the restricted holonomy algebra through transported curvature endomorphisms.	Makes holonomy a derived invariant and global obstruction, not a generic selector.
Shannon chain rule	$H(X) = H(Q) + H(X Q)$ for deterministic Q .	Supplies an invariant decomposition; it does not break fibre automorphism symmetry.
Petz/Grasselli–Streater modules	Quantum monotonicity gives a metric family; mixture/exponential duality uniquely selects BKM up to scale in its stated finite faithful domain.	Quantum connection closure begins only after a metric member is selected.

3 Regular observable strata

Definition 3.1 (Regular observable stratum). *A regular observable stratum is a connected smooth manifold $S \subset Q_{\mathcal{E}}$ on which the declared local metric g is smooth and positive definite and all discrete protocol invariants, including the rank of the characterized projector P , are locally constant.*

Assumption 3.2 (Torsion-free observable transport). *The observable tangent transport introduces no independent infinitesimal translational defect beyond the Lie bracket of observable vector fields:*

$$\text{Tor}^{\nabla}(X, Y) = \nabla_X Y - \nabla_Y X - [X, Y] = 0.$$

This is an explicit axiom, not a consequence of quotient factorization. It is the exact additional condition needed with metric compatibility to eliminate affine connection freedom on TS .

Theorem 3.3 (Regular observable Levi–Civita closure). *Let (S, g) be a regular observable stratum. There exists a unique affine connection ∇^{LC} on TS satisfying*

$$\nabla^{\text{LC}} g = 0, \quad \text{Tor}^{\nabla^{\text{LC}}} = 0.$$

In local coordinates,

$$\Gamma^k_{ij} = \frac{1}{2} g^{k\ell} (\partial_i g_{\ell j} + \partial_j g_{i\ell} - \partial_{\ell} g_{ij}).$$

Proof. Existence follows from the displayed Christoffel formula. Metric compatibility and symmetry in the lower indices are immediate. For uniqueness, the Koszul formula expresses $2g(\nabla_X Y, Z)$ entirely in terms of g , Lie brackets, and directional derivatives of g . Nondegeneracy of g determines $\nabla_X Y$ uniquely. $\square$

Corollary 3.4 (No remaining local tangent-connection representative). *Once g and torsion freedom are part of the fixed regular protocol fibre, no additional local tangent-connection choice remains.*

Non-claim 3.5 (Torsion freedom is declared). RG does not infer torsion freedom from observation extensionality. Models with an independently observed torsional defect belong to a different transport module.

4 The maximal admissible sector as a subbundle

Assumption 4.1 (Smooth tangent-sector realization). *On S , the maximal admissible sector is realized as a smooth constant-rank subbundle*

$$W_{\text{obs|adm}} \subset TS,$$

and $P : TS \rightarrow TS$ is its characterized g -orthogonal projector:

$$P^2 = P, \quad P_g^\dagger = P, \quad \text{im } P = W_{\text{obs|adm}}.$$

Remark 4.2. Constant rank and smoothness hold locally away from the declared discriminant when the spectral gap and all admissibility gates remain open. At threshold crossings this hypothesis can fail, which is a genuine boundary rather than a technical omission.

Theorem 4.3 (Canonical induced admissible connection). *For $X \in \Gamma(TS)$ and $s \in \Gamma(W_{\text{obs|adm}})$ define*

$$\boxed{\nabla_X^{W_{\text{obs|adm}}} s = P \nabla_X^{\text{LC}} s.}$$

Then $\nabla^{W_{\text{obs|adm}}}$ is a connection on $W_{\text{obs|adm}}$, is compatible with the restricted metric $g|_{W_{\text{obs|adm}}}$, and is uniquely induced by the bundled data (S, g, P) through orthogonal projection of the Levi-Civita connection.

Proof. Linearity in X , additivity in s , and the Leibniz rule follow from the corresponding properties of ∇^{LC} and $P(fs) = fP(s)$. Since P maps to $W_{\text{obs|adm}}$, the derivative is a section of $W_{\text{obs|adm}}$. For $s, t \in \Gamma(W_{\text{obs|adm}})$, orthogonality gives

$$g(P \nabla_X^{\text{LC}} s, t) = g(\nabla_X^{\text{LC}} s, t),$$

and similarly in the second slot. Hence

$$Xg(s, t) = g(\nabla_X^{W_{\text{obs|adm}}} s, t) + g(s, \nabla_X^{W_{\text{obs|adm}}} t).$$

Finally, both ∇^{LC} and P are fixed by (g, P) , so their composition contains no additional choice. This is uniqueness as the induced connection, not uniqueness among all abstract metric connections on $W_{\text{obs|adm}}$. $\square$

Theorem 4.4 (Naturality). *Let $f : (S, g, P) \rightarrow (S', g', P')$ be an isometry satisfying $f_* P = P' f_*$. Then*

$$f_*(\nabla_X^{W_{\text{obs|adm}}} s) = \nabla_{f_* X}^{W'_{\text{obs|adm}}}(f_* s).$$

Proof. Levi-Civita connections are natural under isometries. Commutation with the projectors then gives

$$f_* P \nabla_X^{\text{LC}} s = P' f_* \nabla_X^{\text{LC}} s = P' \nabla_{f_* X}^{\text{LC}'} f_* s.$$

$\square$

5 Second fundamental form and sector preservation

Definition 5.1 (Observable second fundamental form). *For $s \in \Gamma(W_{\text{obs|adm}})$ define*

$$\Pi(X, s) = (I - P)\nabla_X^{\text{LC}} s.$$

Theorem 5.2 (Canonical Gauss decomposition). *For every $s \in \Gamma(W_{\text{obs|adm}})$,*

$$\nabla_X^{\text{LC}} s = \nabla_X^{W_{\text{obs|adm}}} s + \Pi(X, s),$$

where the two terms are g -orthogonal. Moreover, the following are equivalent:

1. $\nabla^{\text{LC}} P = 0$;
2. $W_{\text{obs|adm}}$ and $W_{\text{obs|adm}}^\perp$ are parallel subbundles;
3. $\Pi = 0$ and the analogous complementary second fundamental form vanishes;
4. Levi-Civita parallel transport preserves the resolved/null splitting.

Proof. The decomposition is $I = P + (I - P)$. Orthogonality follows from P being g -orthogonal. If $\nabla P = 0$, differentiating $Ps = s$ gives $P\nabla s = \nabla s$, so $W_{\text{obs|adm}}$ is parallel; similarly for the complement. Conversely, preservation of both summands implies that differentiation commutes with the splitting and hence with P . $\square$

Corollary 5.3 (Higher-order closure). *On a regular tangent-sector fibre, the pair*

$$(\nabla^{W_{\text{obs|adm}}}, \Pi)$$

is determined by (g, P) . Thus the first extrinsic higher-order observable datum is no longer arbitrary.

6 Entropy cannot select the connection

Theorem 6.1 (Fibre-entropy invariance no-go). *Let $Q = \pi(X)$ be a deterministic observational quotient of a discrete random variable. Let T be a measure-preserving bijection satisfying $\pi \circ T = \pi$. Then*

$$H(TX) = H(X), \quad H(\pi(TX)) = H(Q),$$

and

$$H(TX \mid \pi(TX)) = H(X \mid Q).$$

Consequently, no objective depending only on the entropy triple

$$(H(X), H(Q), H(X \mid Q))$$

can distinguish elements of the fibre-automorphism group.

Proof. A measure-preserving bijection only relabels atoms, so it preserves the probability multiset and Shannon entropy. Quotient compatibility gives $\pi(TX) = \pi(X)$ almost surely. Conditional entropy invariance follows either directly on each fibre or from the chain rule. $\square$

Corollary 6.2 (Correct role of entropy). *Entropy compatibility can reject transports that change the declared quotient law or fibre probability structure, but cannot choose a representative inside an entropy-preserving stabilizer orbit.*

7 Curvature, holonomy, and global transport

The induced connection has curvature

$$R^{W_{\text{obs|adm}}}(X, Y)s = \nabla_X^{W_{\text{obs|adm}}} \nabla_Y^{W_{\text{obs|adm}}} s - \nabla_Y^{W_{\text{obs|adm}}} \nabla_X^{W_{\text{obs|adm}}} s - \nabla_{[X, Y]}^{W_{\text{obs|adm}}} s.$$

Its parallel transport around loops produces $\text{Hol}(\nabla^{W_{\text{obs|adm}}})$. Ambrose–Singer relates the restricted holonomy algebra to transported curvature endomorphisms. This makes curvature and holonomy canonical diagnostics once $\nabla^{W_{\text{obs|adm}}}$ is fixed.

Non-claim 7.1 (Holonomy is not a generic selector). Prescribing “minimum holonomy” or flatness does not generally characterize a unique connection. Flat connections can possess nontrivial global monodromy on non-simply-connected spaces. In the present theorem ladder, curvature and holonomy are downstream invariants.

8 Rank-changing and non-tangent boundaries

Proposition 8.1 (Rank-change obstruction). *If rank P changes at a point, $\text{im } P$ is not an ordinary smooth vector subbundle in any neighbourhood containing that point. Therefore the induced smooth connection theorem does not cross the rank-change point without a stratified or singular extension.*

Proof. The rank of a smooth vector-bundle projector is locally constant because it equals its trace, a continuous integer-valued function. A rank change contradicts local constancy. $\square$

Non-claim 8.2 (Arbitrary vector bundles). For a general metric vector bundle $E \rightarrow S$ not canonically embedded in TS or another already connected ambient bundle, metric compatibility does not select one connection. An embedding/projection rule, principal connection, divergence duality, physical action, or other module is still required.

9 Quantum module

On faithful finite-dimensional quantum state manifolds, CPTP monotonicity admits the Petz family of monotone metrics rather than one metric. Therefore the general quantum output is a family

$$\{\nabla^{\text{LC}, f}, \nabla^{W_{\text{obs|adm}}, f}\}_{f \in \mathcal{F}_{\text{Petz}}}.$$

If an independent duality axiom requires the mixture and exponential connections to be dual, the Grasselli–Streater theorem selects the BKM metric up to scale in its stated domain. Only then does the same induced-connection closure produce a distinguished BKM member.

Non-claim 9.1 (No unrestricted quantum closure). Quantum connection selection is closed only relative to a selected or characterized quantum metric module. Monotonicity alone does not choose BKM, SLD/Bures, or another Petz metric.

10 Final numerical attack

The deterministic battery used seed 20260816. It sampled a nonconstant positive metric on a two-dimensional regular stratum and a nonparallel rank-one g -orthogonal projector field. Finite differences used a central step 2×10^{-6} .

The nonzero second fundamental form demonstrates that projection is active rather than a trivial restriction. The coordinate-change test transported g , P , vector fields, and sections through a nonsingular affine transformation and recovered the induced derivative within finite-difference error. The entropy attack exhaustively enumerated $S_3 \times S_3$. The rank attack used a

Table 2: Final induced-connection battery.

Test	Maximum residual / result
Levi–Civita metric compatibility, 200 samples	1.11×10^{-16}
Torsion freedom	0
Projector idempotence	2.26×10^{-16}
Projector metric self-adjointness	2.36×10^{-16}
Induced-bundle metric compatibility, 150 samples	1.03×10^{-10}
Gauss decomposition	2.86×10^{-17}
Affine-coordinate naturality, 120 samples	1.98×10^{-10}
Parallel-sector restriction	0
Median nonparallel $\ \mathbf{II}\ $	9.85×10^{-2}
Fibre automorphisms exhausted	36
Ranges of $H(Q)$ and $H(X \mid Q)$	1.11×10^{-16} each
Threshold rank attack	ranks 1 and 2, crossing at $ t = 0.2$

spectral family with eigenvalues $(1, t^2)$ and threshold 0.04, explicitly locating the failure of the smooth constant-rank hypothesis.

Remark 10.1. The computation does not prove the classical theorems. It checks the implementation, attacks covariance and decomposition identities, demonstrates a genuinely bending sector, confirms the entropy stabilizer obstruction, and identifies the exact rank-change boundary.

11 Open-issue ledger

Issue	Status	Exact boundary
Local tangent connection on a regular metric stratum	Conditionally closed	Requires declared metric compatibility and torsion freedom.
Connection on the maximal admissible tangent sector	Conditionally closed	Requires a smooth constant-rank orthogonal subbundle $W_{\text{obs} \text{adm}} \subset TS$.
Entropy as unique connection selector	Closed negatively	Entropy is constant on fibre automorphism orbits.
Higher-order bending datum	Locally closed	$\mathbf{II} = (I - P)\nabla^{\text{LC}}$; further curvature classifications remain.
Global holonomy	Derived but not trivialized	Computable from the induced connection; topology can retain monodromy.
Rank-changing transport	Open	Requires stratified, partial, groupoid, or singular connection theory.
Arbitrary non-tangent vector bundle	Open	Needs an independently characterized ambient connection or extra principle.
Quantum connection	Module-relative	Closed after metric selection; monotonicity alone leaves the Petz family.
Physical transport interpretation	Open	Requires independent dynamical or intervention-level identification.

12 Falsification and revision criteria

The regular closure requires revision if any of the following occurs:

1. the declared metric is degenerate or nonsmooth on the claimed regular stratum;
2. the selected tangent transport is metric compatible and torsion free but differs from Levi–Civita;
3. the reported P is not a smooth orthogonal projector of locally constant rank;
4. $P\nabla^{\text{LC}}$ fails the connection or metric-compatibility identities;
5. an isometry preserving P fails to intertwine the induced connections;
6. entropy alone distinguishes two measure-preserving quotient-fibre automorphisms;
7. a claimed smooth extension crosses a projector rank change without new stratified structure.

13 Conclusion

The connection problem is not solved by adding another arbitrary optimization. It is solved on regular observable strata by recognizing the correct existing geometry. The characterized metric fixes the unique Levi–Civita connection once torsion freedom is declared. The characterized orthogonal projector then induces the canonical admissible-sector connection and second fundamental form:

$$\nabla^{W_{\text{obs|adm}}} = P\nabla^{\text{LC}}, \quad \text{II} = (I - P)\nabla^{\text{LC}}.$$

Entropy remains essential as an invariant and compatibility diagnostic but is mathematically incapable of selecting a representative inside the fibre automorphism group. Curvature and holonomy become derived global invariants. The remaining frontier is now sharply localized at rank-changing discriminants, non-tangent bundles, quantum metric selection, and physical realization.

Reproducibility

The supporting artifacts are:

- `rg_induced_connection_final_battery.py`;
- `rg_induced_connection_final_battery.json`;
- `rg_connection_selection_battery_results.json`;
- `rg_entropy_transport_attack.json`.

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A Battery definitions

The metric used in the attack was

$$g(u, v) = \begin{pmatrix} 1.5 + 0.25u^2 + 0.10v^2 & 0.12uv + 0.04 \sin v \\ 0.12uv + 0.04 \sin v & 1.2 + 0.18v^2 + 0.08u^2 \end{pmatrix}.$$

The rank-one sector was generated by $a(u, v) = (1 + 0.2v, 0.35 + 0.15u)^\top$ and

$$P = \frac{a(a^\top g)}{a^\top g a}.$$

The affine covariance attack used

$$y = Ax + b, \quad A = \begin{pmatrix} 1.3 & 0.25 \\ -0.15 & 0.9 \end{pmatrix}, \quad b = (0.2, -0.1)^\top.$$

The rank-change attack used $L(t) = \text{diag}(1, t^2)$ and $\tau = 0.04$.